

Energy release control for numerical simulations of failure in quasi-brittle solids

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SUMMARY

A path-following constraint is developed which is based on the energy release during failure. This makes it applicable to the simulation of quasi-brittle materials when no previous knowledge is available on the failure behaviour of a body and, consequently, indirect displacement control methods like CMOD cannot be applied. The constraint is derived from the first principle of thermodynamics for a finite-element discretization of a solid with a continuum damage model. The performance of the constraint is demonstrated by means of a bending test on a single-edge-notched beam. Copyright © 2003 John Wiley & Sons, Ltd.

KEY WORDS: path-following technique; arc-length control; quasi-brittle solids; energy release; finite-element method; non-linear analysis

1. INTRODUCTION

One of the fundamental issues in computational analysis of non-linear solids is the development of robust algorithms to trace the equilibrium path properly. The essence of these path-following algorithms consists of coupling the load increment to the increment of selected degrees of freedom through a constraint function [1–3].

When solids are studied which exhibit material instabilities it is often necessary to select the path-following constraint in such a way that only the degrees of freedom involved in the failure process are coupled to the load increment [4]. An example of this kind of constraint is the crack-mouth opening displacement (CMOD) control method, in which the constraint function plays a role analogous to that of the loading servo-control in experimental set-ups. While this kind of constraint is robust, it is not applicable when the location of the failure process zone is not known *a priori*. This is the case, for instance, when Monte-Carlo simulations of heterogeneous materials are carried out. Indeed, the position of the failure process zone is then often determined by the weak spots in the material. The same kind of difficulties

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is found when the location of the failure process zone changes during the failure process. Some methods have been proposed to avoid this difficulty [5, 6]. These methods are based on coupling the load increment to internal variables like the equivalent strain in the elements that belong to the localization zone. The algorithmic cost is, however, increased.

This paper deals with the formulation and implementation of a simple path-following constraint that does not require *a priori* knowledge on the position and behaviour of the failure process zone. This constraint is based on the dissipated energy during the failure process. The performance of the proposed constraint is demonstrated on the problem that motivated the development of the CMOD method, i.e. a single-edge-notched beam loaded in four-point shear [4].

2. TRADITIONAL PATH-FOLLOWING CONSTRAINTS

Consider a finite-element discretization of a solid Ω in which material instabilities can take place. The displacement field $\mathbf{u}$ is then represented by a vector $\mathbf{a}$ of nodal displacements and a matrix $\mathbf{N}$ of shape functions according to the expression

$$\mathbf{u} = \mathbf{N}\mathbf{a} \quad (1)$$

The solid must satisfy the equilibrium equation

$$\mathbf{f}^{\text{int}} = \lambda \hat{\mathbf{f}} \quad (2)$$

in which the essential boundary conditions are prescribed through any suitable method. In this equation, $\mathbf{f}^{\text{int}}$ represents a vector of nodal internal forces. The parameter λ is a loading factor and $\hat{\mathbf{f}}$ is a prescribed nodal loading pattern. The internal force vector is related to the nodal displacement vector $\mathbf{a}$ through a continuum damage model [7]

$$\begin{aligned} \mathbf{f}^{\text{int}} &= \int_{\Omega} \mathbf{B}^T (1 - \omega) \mathbf{D} \boldsymbol{\varepsilon} \, d\Omega \\ &= \int_{\Omega} \mathbf{B}^T (1 - \omega) \mathbf{D} \mathbf{B} \mathbf{a} \, d\Omega \end{aligned} \quad (3)$$

where geometrical linearity has been assumed, $\mathbf{B}$ is the strain-nodal displacement matrix, $\mathbf{D}$ is the elastic constitutive tensor and $0 \leq \omega \leq 1$ is a damage parameter which, in general, depends on an invariant measure of the strain. The solution to (2) is referred to as equilibrium path and is given by the set of processes

$$(\mathbf{a}, \lambda) \quad (4)$$

A parametrization of the equilibrium path at a given point of (4) is considered through the formal expression

$$l(\dot{\mathbf{a}}, \dot{\lambda}; \dot{\tau}) = 0 \quad (5)$$

The equilibrium path (4) is usually computed in an incremental way. An Euler-forward scheme is considered for the parametrization (5), which is formally expressed as

$$\tilde{l}({}^{\tau}\mathbf{a}, {}^{\tau}\lambda, \Delta\mathbf{a}, \Delta\lambda; \Delta\tau) = 0 \quad (6)$$

For each finite increment $\Delta\tau$, the equilibrium path is obtained from the simultaneous solution of Equations (2) and (6)

$$\begin{aligned} \tau + \Delta\tau \mathbf{f}^{\text{int}} - \Delta\lambda \hat{\mathbf{f}} &= \tau \lambda \hat{\mathbf{f}} \\ \tilde{l}(\tau \mathbf{a}, \tau \lambda, \Delta \mathbf{a}, \Delta \lambda; \Delta\tau) &= 0 \end{aligned} \quad (7)$$

When the stress–strain relation does not exhibit descending branches, which situation can be viewed as the absence of material instabilities in a context of geometrical linearity, the parametrization (6) is usually chosen as

$$\Delta\lambda - \Delta\tau = 0 \quad (8)$$

This method is commonly referred to as *force control*. Since τ is a monotonically increasing parameter, Equation (8) means that the loading factor can grow arbitrarily.

When material instabilities occur, the loading factor is not monotonically increasing and different relations must be considered for the parametrization (6). A possibility is the *direct displacement control* method, in which the path parameter is associated to a selected degree of freedom

$$|\tau + \Delta\tau a_i - \tau a_i| - \Delta\tau = 0 \quad (9)$$

such that the selected degree of freedom i corresponds to the only non-zero component of the external force vector $\hat{\mathbf{f}}$. However, this constraint equation is only applicable to situations in which the absolute value of degree of freedom i is monotonically increasing. This precludes a proper description of snap-back behaviour.

Alternatively, the *arc-length control* method can be used. The parametrization (6) is defined as

$$(\tau + \Delta\tau \mathbf{a} - \tau \mathbf{a})^T (\tau + \Delta\tau \mathbf{a} - \tau \mathbf{a}) + \psi \Delta\lambda^2 \hat{\mathbf{f}}^T \hat{\mathbf{f}} - \Delta\tau^2 = 0 \quad (10)$$

where τ can directly be associated to the arc parameter of the equilibrium path. Parameter ψ has little influence in the results and is usually taken $\psi = 0$. Constraint (10) makes it possible to describe the equilibrium path properly in several situations, but it fails when strongly localized deformations take place in the solid. In order to remedy this, the *indirect displacement control* method has been devised [4], which consists of a convenient selection of the degrees of freedom which contribute to the path parameter τ in Equation (10)

$$(\tau + \Delta\tau \mathbf{a} - \tau \mathbf{a})^T \mathbf{\Lambda} (\tau + \Delta\tau \mathbf{a} - \tau \mathbf{a}) - \Delta\tau^2 = 0 \quad (11)$$

where parameter $\psi = 0$ and $\mathbf{\Lambda}$ is a selection matrix. A comparable algorithm has been proposed in [8]. Constraint (11) exhibits great robustness, even when highly localized failure is observed. It makes also possible to simulate the servo-controlled loading that is often used in experimental set-ups. A major disadvantage is that the contributing degrees of freedom must *a priori* be selected. If the failure process zone cannot be predicted by the analyst before carrying out the calculations the method can fail because it cannot be ensured that $\mathbf{\Lambda}$ provides a monotonically increasing function of the degrees of freedom in (11). In the next section, an alternative formulation is derived that is based on the released energy, which is a monotonically increasing quantity during failure. For consistency with the methods described above, this method will be referred to as *energy release control*.

3. PATH-FOLLOWING CONSTRAINT BASED ON ENERGY RELEASE

The path parameter τ introduced in (5) must be related to a monotonically increasing variable of the solid. A physically appealing possibility is provided by the energy release, which always exhibits a positive rate when failure evolves in quasi-brittle materials.

The energy release rate during failure is obtained from the first principle of thermodynamics. In a context of quasi-static loading this is written as [7]

$$\dot{V} = P - G \quad (12)$$

where $\dot{V}$ is the elastic potential energy rate, P is the power of the external forces and G is the energy release rate. The elastic potential energy V stored in the discretized solid at an equilibrium configuration satisfying Equation (2) is

$$\begin{aligned} V &= \frac{1}{2} \int_{\Omega} \varepsilon^T \sigma \, d\Omega = \frac{1}{2} \int_{\Omega} \mathbf{a}^T \mathbf{B}^T \sigma \, d\Omega \\ &= \frac{1}{2} \mathbf{a}^T \mathbf{f}^{\text{int}} = \frac{1}{2} \mathbf{a}^T \lambda \hat{\mathbf{f}} \end{aligned} \quad (13)$$

The elastic potential energy rate is obtained by differentiating (13)

$$\dot{V} = \frac{1}{2} \dot{\mathbf{a}}^T \lambda \hat{\mathbf{f}} + \frac{1}{2} \mathbf{a}^T \dot{\lambda} \hat{\mathbf{f}} \quad (14)$$

The power of the external forces is expressed as

$$P = \dot{\mathbf{a}}^T \lambda \hat{\mathbf{f}} \quad (15)$$

After substitution of (14) and (15) into (12), the expression

$$G = \frac{1}{2} (\lambda \dot{\mathbf{a}}^T - \dot{\lambda} \mathbf{a}^T) \hat{\mathbf{f}} \quad (16)$$

results for the energy release rate. Equation (16) can be used in a particular form of parametrization (5) as

$$l(\dot{\mathbf{a}}, \dot{\lambda}; \dot{\tau}) = \frac{1}{2} (\lambda \dot{\mathbf{a}}^T - \dot{\lambda} \mathbf{a}^T) \hat{\mathbf{f}} - \dot{\tau} \quad (17)$$

An Euler-forward integration scheme for this parametrization is then written as

$$\frac{1}{2} [\tau \lambda (\tau + \Delta\tau \mathbf{a}^T - \tau \mathbf{a}^T) - \Delta\lambda \tau \mathbf{a}^T] \hat{\mathbf{f}} - \Delta\tau = 0 \quad (18)$$

The area enclosed by the shaded triangle in Figure 1 provides a geometrical interpretation of this definition of $\Delta\tau$.

4. SOLUTION PROCEDURE

The equilibrium path is traced by solving the equilibrium equation (2) together with constraint (18), as in Reference [1]. Using the notation

$$\mathbf{K}_s = \int_{\Omega} \mathbf{B}^T (1 - \omega) \mathbf{D} \mathbf{B} \, d\Omega \quad (19)$$

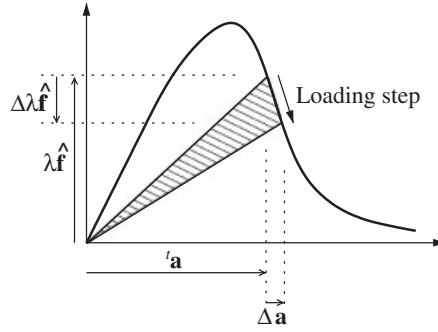


Figure 1. Geometrical interpretation of the definition of $\Delta\tau$ according to Equation (18). The shaded area corresponds to the energy release during a loading step.

for the secant stiffness matrix and assuming that a point $({}^\tau\mathbf{a}, {}^\tau\lambda)$ of the equilibrium path has been reached, a vector of residuals is defined as

$$\mathbf{r}({}^{\tau+\Delta\tau}\mathbf{a}, \Delta\lambda) = \begin{pmatrix} \mathbf{K}_s{}^{\tau+\Delta\tau}\mathbf{a} - \Delta\lambda\hat{\mathbf{f}} - {}^\tau\lambda\hat{\mathbf{f}} \\ \frac{1}{2}[\lambda({}^{\tau+\Delta\tau}\mathbf{a}^T - {}^\tau\mathbf{a}^T) - \Delta\lambda{}^\tau\mathbf{a}^T]\hat{\mathbf{f}} - \Delta\tau \end{pmatrix} \quad (20)$$

The point $({}^{\tau+\Delta\tau}\mathbf{a}, {}^{\tau+\Delta\tau}\lambda)$ of the equilibrium path is then computed by solving the equation

$$\mathbf{r}({}^{\tau+\Delta\tau}\mathbf{a}, \Delta\lambda) = 0 \quad (21)$$

through the Newton–Raphson procedure

$$\begin{pmatrix} {}^{\tau+\Delta\tau}\mathbf{a} \\ \Delta\lambda \end{pmatrix}^{(k+1)} = \begin{pmatrix} {}^{\tau+\Delta\tau}\mathbf{a} \\ \Delta\lambda \end{pmatrix}^{(k)} - \left[\frac{\partial \mathbf{r}}{\partial ({}^{\tau+\Delta\tau}\mathbf{a}, \Delta\lambda)} \right]^{-1} \mathbf{r}^{(k)} \quad (22)$$

where k represents the iteration number and the Jacobian matrix is of the form

$$\frac{\partial \mathbf{r}}{\partial ({}^{\tau+\Delta\tau}\mathbf{a}, \Delta\lambda)} = \begin{bmatrix} \frac{\partial \mathbf{K}_s}{\partial \mathbf{a}}{}^{\tau+\Delta\tau}\mathbf{a} + \mathbf{K}_s & -\hat{\mathbf{f}} \\ \frac{1}{2}{}^\tau\lambda\hat{\mathbf{f}}^T & -\frac{1}{2}{}^\tau\mathbf{a}^T\hat{\mathbf{f}} \end{bmatrix} \quad (23)$$

Procedure (22) is iterated until convergence upon a suitable norm is attained. The loading factor is then updated as

$${}^{\tau+\Delta\tau}\lambda = {}^\tau\lambda + \Delta\lambda \quad (24)$$

5. STEP-SIZE ESTIMATION

The parametrization given by (18) is related to the energy release and can only be used when damage is evolving. This is, in general, not the case at the initial stages of the loading

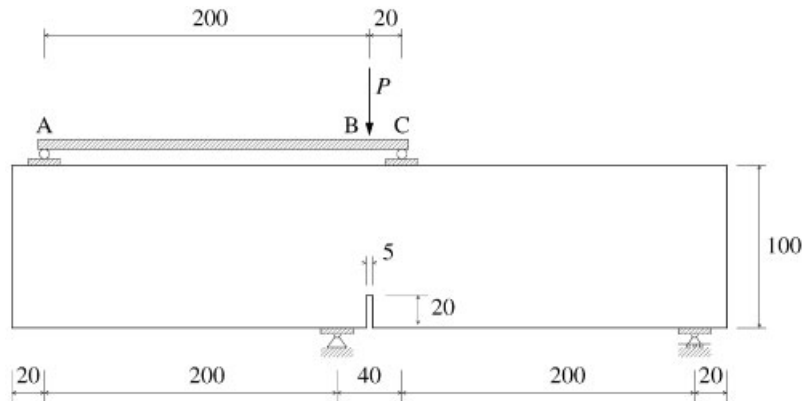


Figure 2. Schematic representation of a four-point shear loading test on a single-edge-notched concrete beam (dimensions in millimetres).

process, in which the whole body is still in the elastic regime. The adopted procedure is then to start the loading process with load control, cf. Equation (8). When the degree of non-linearity is sufficiently increased, one can switch from load control to energy release control. The degree of non-linearity is measured in an informal fashion by the number of required Newton–Raphson iterations in (22), as proposed in [9]. If the optimal number of iterations is given by n_d (usually 5), it is assumed that the degree of non-linearity is sufficiently increased when the number of iterations n_i required to achieve convergence at load step i is larger than n_d . The energy release during this last step i is then computed from (18) and it provides an initial estimate for the step size $\Delta\tau$. In subsequent steps the step size is adapted according to

$$\Delta\tau_{j+1} = \Delta\tau_j \frac{n_d}{n_j} \quad (25)$$

This step-size estimation method works properly and keeps much resemblance with the technique proposed in Reference [5], in which the new load increment at the first iteration of each step is related to the number of iterations required in the previous step as well as to a quantity related to the work of the external forces.

6. APPLICATION EXAMPLE

The performance of the energy release control method is demonstrated by means of a single-edge-notched (SEN) concrete beam subjected to a skew-symmetric four-point shear loading. This example was also used in Reference [4] to demonstrate the indirect displacement control method. The static scheme of the beam, which has a thickness of 100 mm, is shown in Figure 2.

A gradient enhanced damage model is used [10]. The damage parameter introduced in Equation (3) is a function of a history parameter κ which is related to a spatially averaged

measure $\bar{\varepsilon}^{\text{eq}}$ of the equivalent strain through the Kuhn–Tucker conditions

$$\dot{\kappa} \geq 0, \quad \bar{\varepsilon}^{\text{eq}} - \kappa \leq 0, \quad \dot{\kappa}(\bar{\varepsilon}^{\text{eq}} - \kappa) = 0 \quad (26)$$

The averaged measure $\bar{\varepsilon}^{\text{eq}}$ is obtained from the diffusion equation

$$\bar{\varepsilon}^{\text{eq}} - c \nabla^2 \bar{\varepsilon}^{\text{eq}} = \varepsilon^{\text{eq}} \quad (27)$$

where parameter c is of dimension length squared and quantifies the size of the zone in which damage evolves and the equivalent strain ε^{eq} follows from

$$\varepsilon^{\text{eq}} = \frac{\xi - 1}{2\xi(1 - 2\nu)} I_1 + \frac{1}{2\xi} \sqrt{\frac{(\xi - 1)^2}{(1 - 2\nu)^2} I_1^2 + \frac{2\xi}{(1 + \nu)^2} J_2} \quad (28)$$

where the strain tensor invariants are given by

$$\begin{aligned} I_1 &= \varepsilon_1 + \varepsilon_2 + \varepsilon_3 \\ J_2 &= (\varepsilon_1 - \varepsilon_2)^2 + (\varepsilon_2 - \varepsilon_3)^2 + (\varepsilon_3 - \varepsilon_1)^2 \end{aligned} \quad (29)$$

and the parameter ξ controls the sensitivity to tension with respect to that in compression. This definition of the equivalent strain was introduced in Reference [11]. The damage parameter obeys the exponential evolution law

$$\omega(\kappa, \kappa_0) = \begin{cases} 1 - \frac{\kappa_0}{\kappa} [(1 - \alpha) + \alpha \exp(-\beta(\kappa - \kappa_0))] & \text{if } \kappa > \kappa_0 \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

where α and β are parameters that determine the strength decay and κ_0 represents the threshold for damage initiation.

In the simulations presented in Reference [10] the Young's modulus is $E = 35$ GPa, the Poisson's ratio is $\nu = 0.2$, the gradient parameter in Equation (27) is $c = 1 \text{ mm}^2$, the parameters α and β of Equation (30) are 0.96 and 100, respectively, the sensitivity to tension with respect to that in compression in Equation (28) is $\xi = 10$ and the threshold for damage initiation is $\kappa_0 = 6 \times 10^{-5}$. These material parameters are chosen in order to reproduce the experiments presented in Reference [12] on normal weight concrete. In the simulations presented here, the beam is discretized into eight-noded plane-stress finite elements with a 2×2 Gauss–Legendre integration quadrature. The loading platens are modelled by means of suitable linear constraints and the beam ABC is assumed to be rigid. A reference load $P = 1.1$ kN is considered. The initial load factor has been set to 4, the maximal load factor has been limited to 10 and the termination load factor is 4. Convergence is checked on the norm of the residual $\mathbf{r}^{(k)}$ defined in Equation (20) relative to that of $\mathbf{r}^{(0)}$, with a tolerance 1×10^{-5} .

The load displacement curves for points A, B and C are represented in Figure 3 and the number of iterations required at each loading step is represented in Figure 4. Unfortunately, the behaviour of the normal weight SEN-beam in both the experiments and the numerical simulations is too 'mild' and the severe snap-backs observed in Reference [4] are not found. In order to accentuate the snap-back behaviour and to demonstrate the performance of the energy release control method, the simulations have also been carried out with lightweight concrete. For this purpose, the Young's modulus has been taken $E = 18$ GPa, parameter $\beta = 350$ and

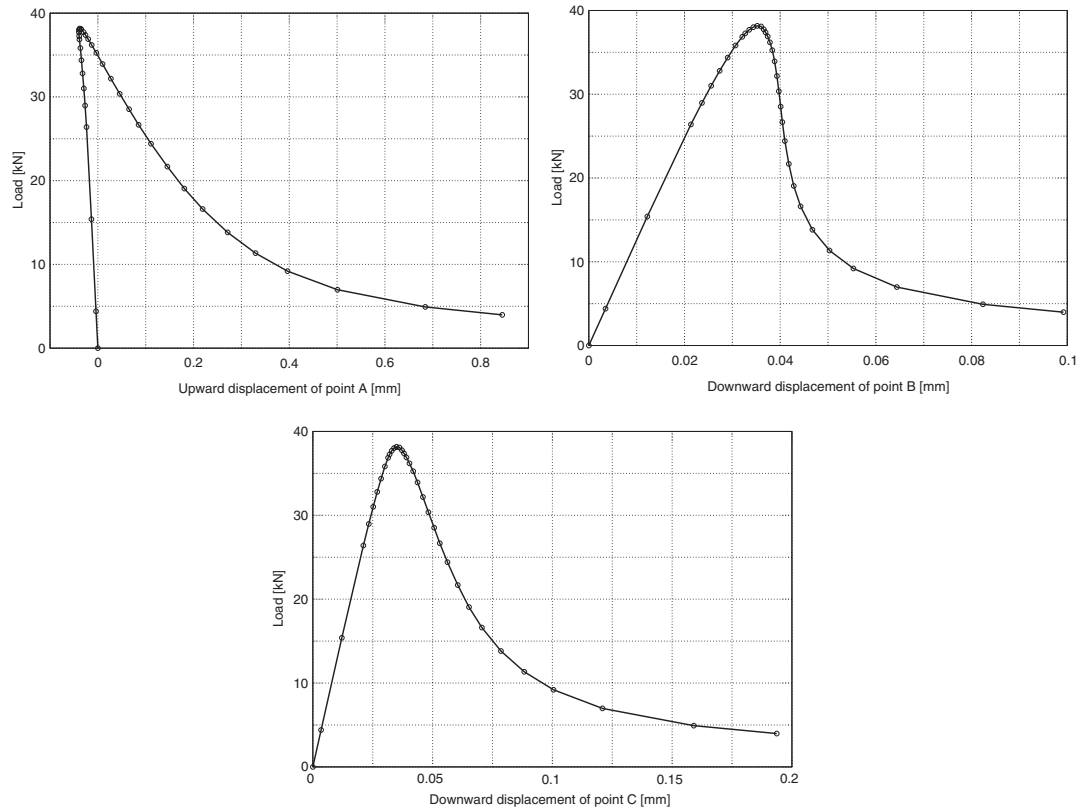


Figure 3. SEN-beam of normal weight concrete: Load displacement curves for points A, B and C.

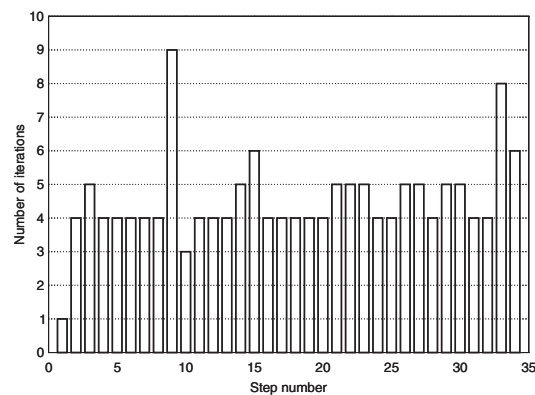


Figure 4. SEN-beam of normal weight concrete: number of iterations required to attain convergence at each loading step.

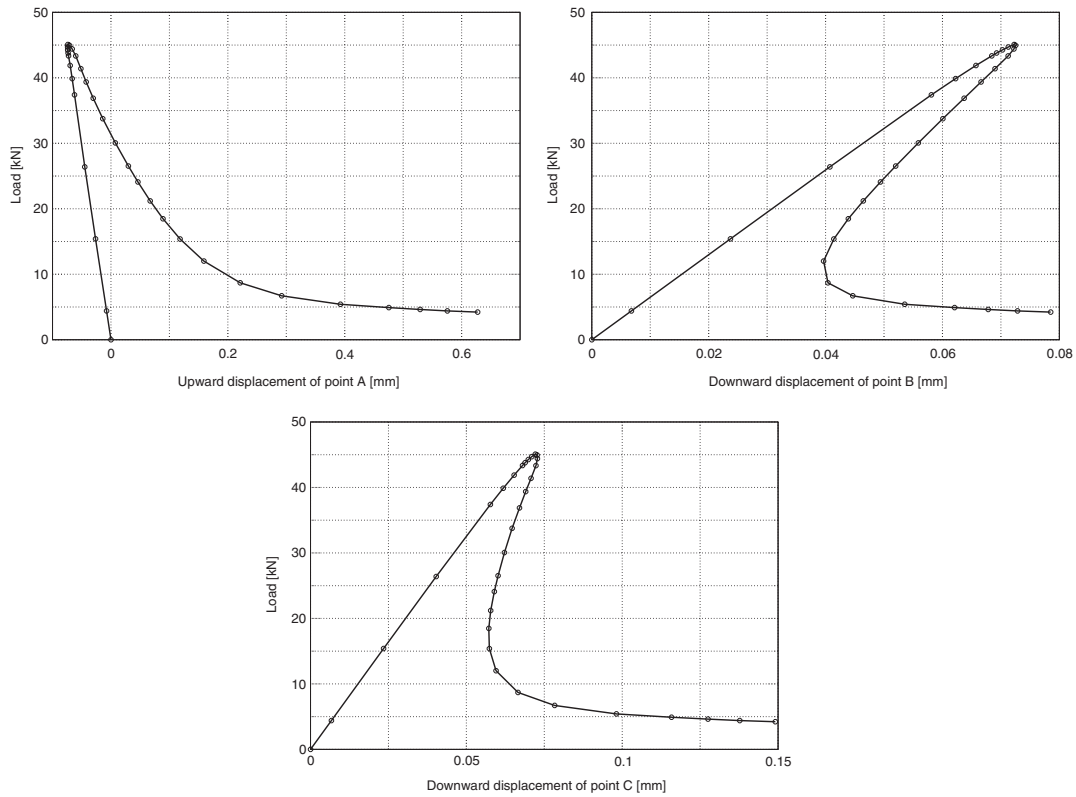


Figure 5. SEN-beam of lightweight concrete: load displacement curves for points A, B and C.

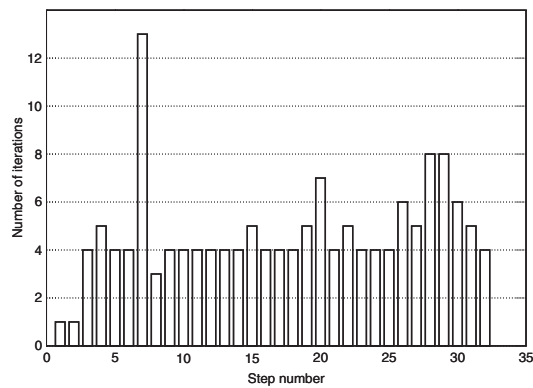


Figure 6. SEN-beam of lightweight concrete: number of iterations required to attain convergence at each loading step.

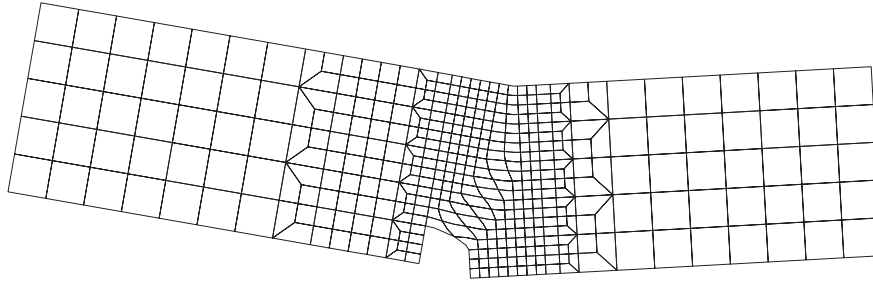


Figure 7. SEN-beam of lightweight concrete: nodal displacement field at the final load step.

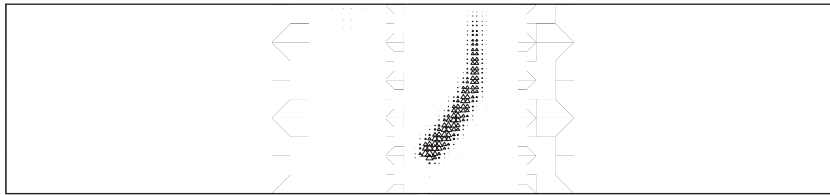


Figure 8. SEN-beam of lightweight concrete: profile of equivalent strain in the damaged zone at the final load step.

$\kappa_0 = 2.1 \times 10^{-4}$. These values are in accordance with those used in Reference [10] to simulate tensile failure in double-edge-notched concrete specimens and lead to a more brittle response. The expected snap-back behaviour is demonstrated in Figure 5 and the required number of iterations is shown in Figure 6. The nodal displacement field and the equivalent strain measure in the damaged zone are represented in Figures 7 and 8, respectively, at the final load step.

The performance of the step-size estimation algorithm (25) is worthy of mention. At load step 9 of the normal weight concrete beam and at load step 7 of the lightweight concrete beam nine iterations and 13 iterations are required, respectively. The good convergence properties of the previous steps have led to a too large increment for the path parameter $\Delta\tau$. The step size is then drastically reduced in the subsequent step, as can be observed from Figures 3 and 5 and the convergence properties are consequently improved, cf. Figures 4 and 6.

7. CLOSURE

A path-following constraint has been formulated which depends on the energy release. Since this energy is always an increasing variable during a failure process, no *a priori* knowledge is required on the failure pattern of the structure. This makes this constraint especially applicable to problems which exhibit a severe snap behaviour or for which the location of the failure process zone is difficult to predict *a priori*. The performance of the constraint is shown on a single-edge-notched beam of both normal weight and lightweight concrete.

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